(3) Ch-3

🎓 UNIT II — LABOUR DEMAND (PART 1)

SECTION 3-1: THE PRODUCTION FUNCTION

📜 THESIS STATEMENT:

> **Labor demand is a "derived demand" determined by the firm's technology and the consumer's desire for the final product.** The production function describes the technological constraints facing the firm, specifically the **Law of Diminishing Returns**, which dictates that adding more labor to a fixed capital stock eventually yields progressively smaller gains in output.

A. THE MODEL: TECHNOLOGY

The firm combines inputs to create output (q).

- **Production Function:** $q = f(E, K)$
  - E: Employee-hours (Labor).
  - K: Capital (Land, machines, buildings).
- **Assumption:** Labor is homogeneous (all workers are identical) for the basic model.

B. KEY PRODUCTIVITY CONCEPTS

1. Marginal Product of Labor (MP_E)

- **Definition:** The change in output resulting from hiring one additional worker, **holding capital constant**.

$$MP_E = \frac{\Delta q}{\Delta E}$$

- **Slope:** MP_E is the slope of the Total Product curve.

2. Average Product of Labor (AP_E)

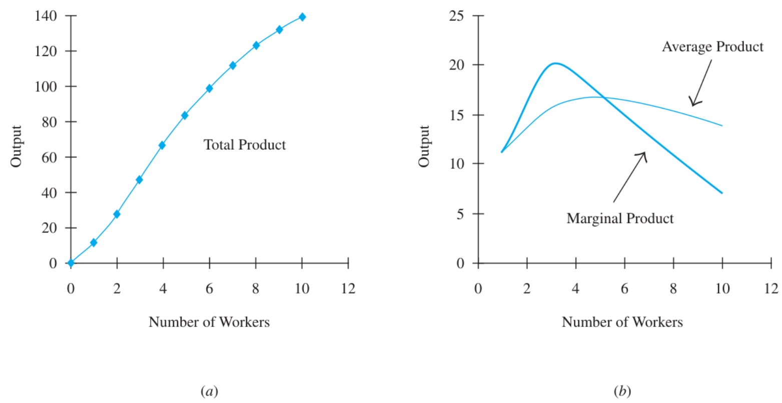
- **Definition:** The amount of output produced by the "typical" worker.

$$AP_E = \frac{q}{E}$$

3. The Law of Diminishing Returns

- **The Mechanism:** Initially, adding workers to fixed capital might increase productivity due to specialization (increasing returns). However, eventually, the facility becomes crowded, and capital is shared too thinly.
- **Result:** MP_E must eventually decline.
- **Implication:** If diminishing returns did not exist, firms would expand indefinitely.

FIGURE 3-1 **The Total Product, the Marginal Product, and the Average Product Curves**
(*a*) The total product curve gives the relationship between output and the number of workers hired by the firm (holding capital fixed). (*b*) The marginal product curve gives the output produced by each additional worker, and the average product curve gives the output per worker.



> [!quote] Geometric Rule: > * If $MP_E > AP_E$, the Average Product is **rising**. > * If $MP_E < AP_E$, the Average Product is **falling**. > * Therefore, the MP_E curve intersects the AP_E curve at the **maximum point** of the

Average Product curve.

SECTION 3-2: THE EMPLOYMENT DECISION IN THE SHORT RUN

📜 THESIS STATEMENT:

> **In the short run (fixed capital), a profit-maximizing firm hires workers up to the point where the value of the worker's output equals the cost of hiring them.** Consequently, the **Value of Marginal Product (VMP_E)** curve constitutes the firm's short-run demand curve for labor.

A. THE OPTIMIZATION PROBLEM

- **Short Run Definition:** Capital (K) is fixed at level K_0. The firm can only adjust Labor (E).
- **Firm's Objective:** Maximize Profit.

$$\text{Profit} = pq - wE - rK$$

  - p: Output price (Constant / Perfect Competition).
  - w: Wage rate (Constant).
  - r: Price of capital.

B. THE HIRING RULE (Marginal Productivity Condition)

The firm compares the cost of hiring a worker against the revenue that worker generates.

1. **Value of Marginal Product (VMP_E):**
   The dollar value of the extra output produced by the last worker.

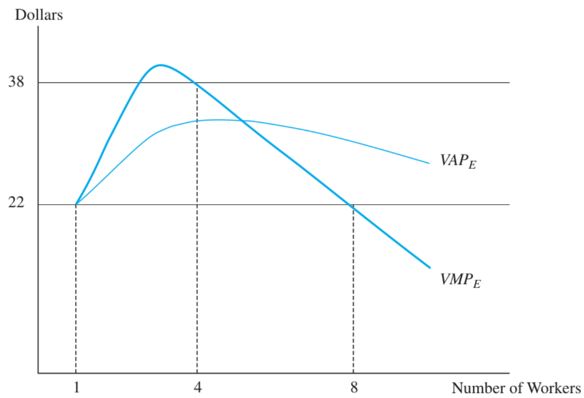
$$VMP_E = p \times MP_E$$

2. **The Decision Logic:**
   - If $VMP_E > w$: The worker brings in more revenue than they cost. **HIRE.**
   - If $VMP_E < w$: The worker costs more than they generate. **DO NOT HIRE / FIRE.**
3. **The Equilibrium Condition:**
   The firm stops hiring exactly when:

$$VMP_E = w$$

   - *Constraint:* This intersection must occur on the **downward-sloping** portion of the VMP_E curve (diminishing returns).

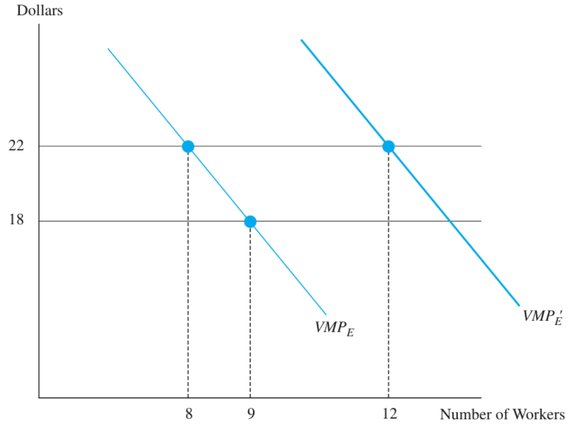
FIGURE 3-2 **The Firm's Hiring Decision in the Short Run**
A profit-maximizing firm hires workers up to the point where the wage rate equals the value of marginal product of labor. If the wage is $22, the firm hires eight workers.



C. THE SHORT-RUN LABOR DEMAND CURVE

- **Derivation:** Because the firm always sets E such that $VMP_E = w$, the downward-sloping portion of the VMP_E curve **IS** the labor demand curve.
- **Slope:** Downward sloping because of Diminishing Marginal Product.
  - Wage $\downarrow \implies$ Firm hires more workers (Movement along curve).
- **Shifts:**
  - **Output Price (p) rises:** VMP_E shifts UP $\implies$ Demand for labor increases.
  - **Technology improves (MP_E rises):** VMP_E shifts UP $\implies$ Demand for labor increases.

FIGURE 3-3 **The Short-Run Demand Curve for Labor**
Because marginal product eventually declines, the short-run demand curve for labor is downward sloping. A drop in the wage from \$22 to \$18 increases the firm's employment. An increase in the price of the output shifts the value of marginal product curve upward and increases employment.



D. EQUIVALENCE TO OUTPUT MARKET RULES

The labor market rule ($w = VMP_E$) is mathematically identical to the output market rule ($p = MC$).

- **Cost of producing 1 unit:** To produce 1 extra unit, you need $1/MP_E$ workers. Each worker costs w.
- **Marginal Cost (MC):** w/MP_E.
- **Output Rule:** Produce where $p = MC$.

$$p = \frac{w}{MP_E} \Rightarrow p \times MP_E = w$$

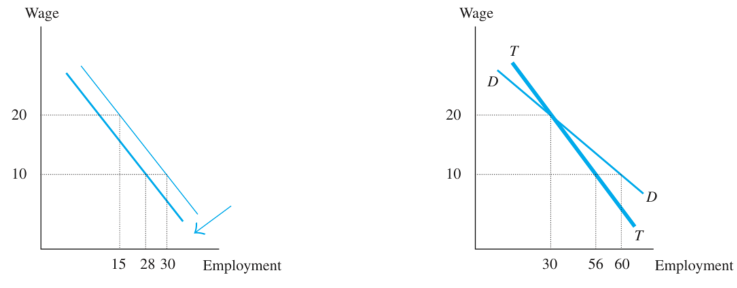
  - *Conclusion:* A firm maximizing profit in the product market is simultaneously maximizing profit in the labor market.

E. AGGREGATION: FROM FIRM TO INDUSTRY DEMAND

Exam Warning: You cannot simply sum individual firm demand curves horizontally to get industry demand.

1. **The Problem:** If the wage falls, *every* firm expands output.
2. **The Feedback Effect:** As total industry output rises, the **market price of the good (p) falls**.
3. **The Adjustment:**
   - Wage falls $\rightarrow$ Firms try to hire more.
   - Output supply surges $\rightarrow$ Output Price (p) falls.
   - Since $VMP_E = p \times MP_E$, the drop in p shifts the individual demand curves **inward (left)**.
4. **Result:** The Industry Demand Curve is **steeper (more inelastic)** than the horizontal sum of individual firm curves.

FIGURE 3-4 **The Short-Run Demand Curve for the Industry**
Each firm in the industry hires 15 workers when the wage is \$20. If the wage falls to \$10, each firm now wants to hire 30 workers. If all firms expand, the supply of the output in the industry increases, reducing the price of the output and reducing the value of marginal product, so the labor demand curve of each individual firm shifts slightly to the left. At the lower price of \$10, each firm then hires only 28 workers. The industry demand curve is not given by the horizontal sum of the firm's demand curves (*DD*), but takes into account the impact of the industry's expansion on output price (*TT*).



(*a*) Individual Firms (*b*) Industry

> [!important] Exam Note: Definitions > * **Short Run:** Period where K is fixed. > * **Elasticity of Labor Demand (δ_{SR}):** $\% \Delta E / \% \Delta w$. > * **Elastic:** $|\delta| > 1$. > * **Inelastic:** $|\delta| < 1$.

F. CRITICISMS OF MARGINAL PRODUCTIVITY THEORY

Borjas raises and rejects a key objection to the entire framework.

- **The Critique:** Firms in the real world do not consciously calculate MP_E or equate it to the wage. The theory seems to demand unrealistic computational ability from employers.

- **The Rebuttal (The Survivor Argument):**
  - We do **not** need to assume that firms literally solve calculus problems.
  - We only need to assume that firms which *behave as if* they follow the $VMP_E = w$ rule will tend to earn higher profits.
  - In a competitive market, firms that systematically deviate from this rule will incur higher costs, earn lower profits, and eventually be **driven out of the market**.
  - **Result:** Only profit-maximizing survivors remain. The market selects for the behavior the theory predicts, regardless of whether any individual manager consciously thinks in these terms.

> Borjas's Point
>
> The marginal productivity theory is a theory about **market outcomes**, not a theory about the psychology of individual managers.

📝 COMPARATIVE TABLE: SHIFTS VS. MOVEMENTS

Scenario	Effect on VMP_E Curve	Effect on Employment (E)
Wage Decrease ($w \downarrow$)	No Shift (Movement Down)	**Increases**
Output Price Rise ($p \uparrow$)	Shifts **Right/Up**	**Increases**
Tech Progress ($MP_E \uparrow$)	Shifts **Right/Up**	**Increases**
Capital Stock Rise ($K \uparrow$)	Shifts **Right/Up** (usually)	**Increases** (if scale effect dominates)

📝 EXAM PREPARATION NOTES

Possible Exam Questions:

1. *Derive the short-run demand curve for labor for a perfectly competitive firm. Why is it downward sloping?* (Answer: Diminishing Marginal Product).
2. *Why is the industry demand curve for labor steeper than the summation of individual firm demand curves?* (Answer: Output price effect).
3. *Show that the condition $w = VMP_E$ is equivalent to the condition $P = MC$.*

Diagram Checklist:

- **Figure 3-1:** Total Product (S-shape) and MP/AP curves (Inverted U-shapes). Show MP cutting AP at the max.
- **Figure 3-2:** The Hiring Decision. Intersection of horizontal Wage line and downward sloping VMP_E.
- **Figure 3-4:** Industry Demand. Show the "naive" summation vs. the "true" steeper curve due to price adjustment.

SECTION 3-3: THE EMPLOYMENT DECISION IN THE LONG RUN

📜 THESIS STATEMENT:

> **In the long run, capital is not fixed.** The firm maximizes profit by simultaneously choosing the optimal level of both Labor (E) and Capital (K). This decision process requires minimizing the cost of producing any given level of output by equating the **marginal productivity per dollar** across all inputs.

A. THE TECHNOLOGY: ISOQUANTS

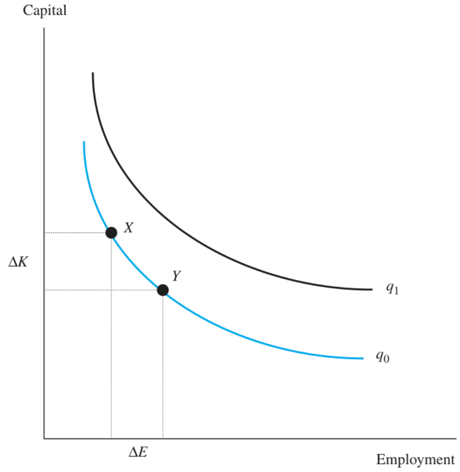
Just as workers have indifference curves, firms have **Isoquants**.

- **Definition:** An isoquant represents all possible combinations of Labor (E) and Capital (K) that produce the **same** level of output (q_0).
- **Properties:**
  1. **Downward Sloping:** To keep output constant, if you use less capital, you must use more labor.
  2. **Convex to Origin:** As you substitute labor for capital, it becomes increasingly difficult to replace the remaining machines

with workers (Diminishing Marginal Rate of Technical Substitution).

3. **Non-intersecting:** Higher isoquants imply higher output.

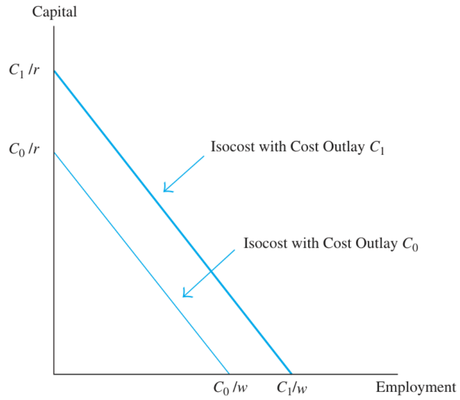
FIGURE 3-6 **Isoquant Curves**
All capital-labor combinations that lie along a single isoquant produce the same level of output. The input combinations at points X and Y produce q_0 units of output. Input combinations that lie on higher isoquants produce more output.



B. THE BUDGET: ISOCOSTS

- **Definition:** An isocost line shows all combinations of E and K that cost the **same** total amount (C).
- **Equation:** $C = wE + rK$
  - Rearranged: $K = (C/r) - (w/r)E$
- **Slope:** The slope is the ratio of input prices: $-w/r$.

FIGURE 3-7 **Isocost Lines**
All capital-labor combinations that lie along a single isocost curve are equally costly. Capital-labor combinations that lie on a higher isocost curve are more costly. The slope of an isoquant equals the ratio of input prices ($-w/r$).



C. COST MINIMIZATION (The Tangency Condition)

To maximize profit, a firm must produce any chosen level of output at the lowest possible cost.

- **The Optimal Point:** Where the Isoquant is **tangent** to the lowest possible Isocost line.
- **The Condition:**

$$\text{Slope of Isoquant} = \text{Slope of Isocost}$$

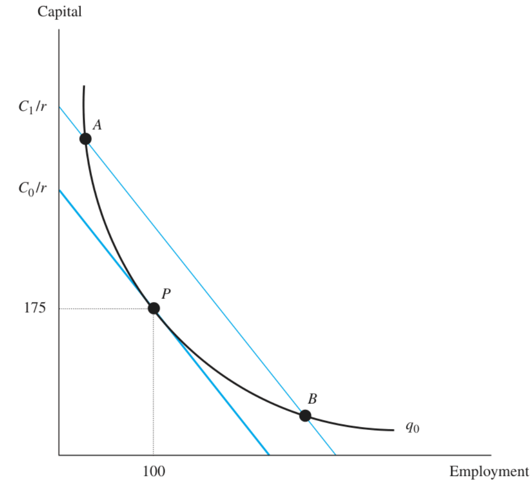
$$\frac{MP_E}{MP_K} = \frac{w}{r}$$

- **Economic Interpretation ("Bang for the Buck"):**
  Rearranging the terms:

$$\frac{MP_E}{w} = \frac{MP_K}{r}$$

> The firm optimizes when the last dollar spent on Labor yields the same amount of extra output as the last dollar spent on Capital.

FIGURE 3-8 **The Firm's Optimal Combination of Inputs**
A firm minimizes the costs of producing q_0 units of output by using the capital-labor combination at point *P*, where the isoquant is tangent to the isocost. All other capital-labor combinations (such as those given by points *A* and *B*) lie on a higher isocost curve.



D. PROFIT MAXIMIZATION VS. COST MINIMIZATION

Borjas makes a crucial distinction here:

1. **Cost Minimization:** Tells the firm the best *ratio* of Labor to Capital for a specific output.
2. **Profit Maximization:** Tells the firm *how much* output to actually produce.
   - **The Rule:** The firm hires both inputs until the input price equals the Value of Marginal Product for **both** inputs:

$$w = p \times MP_E \quad \text{AND} \quad r = p \times MP_K$$

SECTION 3-4: THE LONG-RUN DEMAND CURVE FOR LABOR

📜 THESIS STATEMENT:

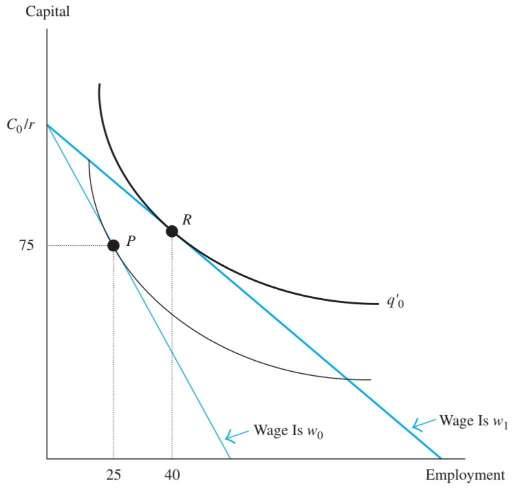
> **The long-run labor demand curve is downward sloping because a wage decline triggers two reinforcing mechanisms: the Substitution Effect (using more labor instead of capital) and the Scale Effect (expanding production because costs have fallen).** Consequently, labor demand is **more elastic** in the long run than in the short run.

THE WRONG APPROACH — Figure 3-9

When the wage falls from w_0 to w_1, the isocost line flattens (its slope is $-w/r$, so a lower w means a less steep line). The **tempting but incorrect** move is to **rotate the isocost around the original vertical intercept** C_0/r, holding total cost outlay constant at C_0.

> **Why is this wrong?** Rotating around C_0/r implicitly assumes the firm spends the same total amount of money before and after the wage change. But there is **nothing in profit-maximization theory** that forces the firm to hold its cost outlay constant. The firm's real constraints are: (1) the production technology, and (2) constant output price p and capital price r. Cost outlay is an *endogenous outcome*, not a fixed constraint.

FIGURE 3-9 **The Impact of a Wage Reduction, Holding Constant Initial Cost Outlay at C_0**
A wage reduction flattens the isocost curve. If the firm were to hold the initial cost outlay constant at C_0 dollars, the isocost would rotate around C_0 and the firm would move from point *P* to point *R*. A profit-maximizing firm, however, will not generally want to hold the cost outlay constant when the wage changes.



> If we incorrectly did this rotation, the firm would move from **P → R (wrong)**, hiring 40 workers and expanding output to q_0'. This analysis is simply invalid.

THE CORRECT APPROACH — Figures 3-10 & 3-11

The correct two-step reasoning is:

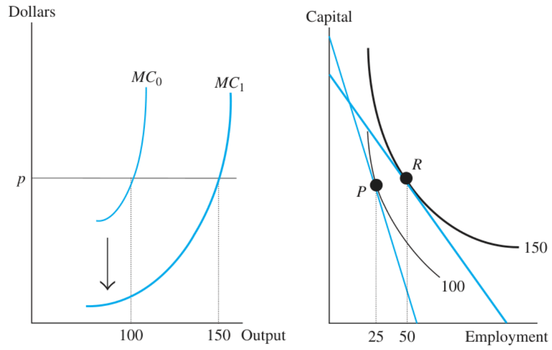
Step 1 — Output Decision (Figure 3-10a):

A wage cut reduces the **Marginal Cost (MC)** of production, because it is now cheaper to produce each additional unit of output when labor is inexpensive. Since profit maximization requires $p = MC$, a fall in MC means the existing output level is no longer optimal — the firm can profitably expand.

$$w \downarrow \implies MC \downarrow \implies MC_1 < MC_0 \implies \text{New profit-max output} > \text{old output}$$

In Figure 3-10a: the MC curve shifts down from MC_0 to MC_1, and the firm expands from **100 to 150 units** of output.

FIGURE 3-10 **The Impact of a Wage Reduction on the Output and Employment of a Profit-Maximizing Firm**
(*a*) A wage cut reduces the marginal cost of production and encourages the firm to expand (from producing 100 to 150 units). (*b*) The firm moves from point *P* to point *R*, increasing the number of workers hired from 25 to 50.



(*a*) Firm's Output Decision (*b*) Firm's Hiring Decision

The firm now moves to a **higher isoquant** (150 units instead of 100). Critically:

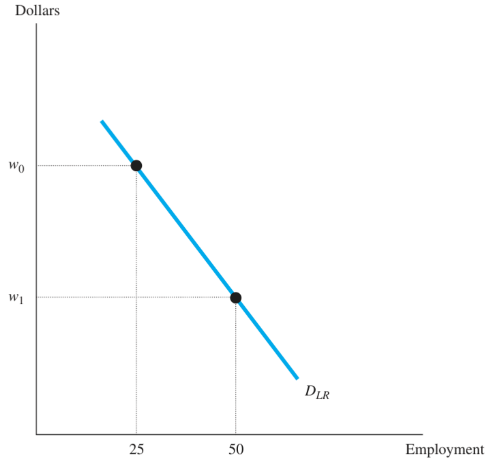
- The new isocost line has a **flatter slope** (reflecting $w_1 < w_0$) but need **not share the same vertical intercept** as the original — total cost outlay has changed.
- The firm produces the new, higher output level **efficiently**, meaning it picks the tangency point of the new (flatter) isocost with the higher isoquant.
- This new optimum is **Point R**.

As drawn, the firm's employment rises from **25 → 50 workers**. Capital may also rise (scale effect on capital), though this is not guaranteed.

THE LONG-RUN DEMAND CURVE — Figure 3-11

Connecting the two equilibria across different wages directly gives the **long-run demand curve for labor (D_{LR})**:

FIGURE 3-11 **The Long-Run Demand Curve for Labor**
The long-run demand curve for labor gives the firm's employment at a given wage and is downward sloping.



The D_{LR} curve is **downward sloping**. This can be proven formally: as long as both labor and capital are **normal inputs** (the firm uses more of both as it expands at constant input prices), the firm will always hire more labor when the wage falls. The long-run demand curve **must therefore be negatively sloped**.

> **Key Contrast:** Note that the new isocost in Figure 3-10b does *not* originate from C_0/r. This is the whole point — the firm is not constrained to spend C_0. It is free to expand its cost outlay as profits dictate.

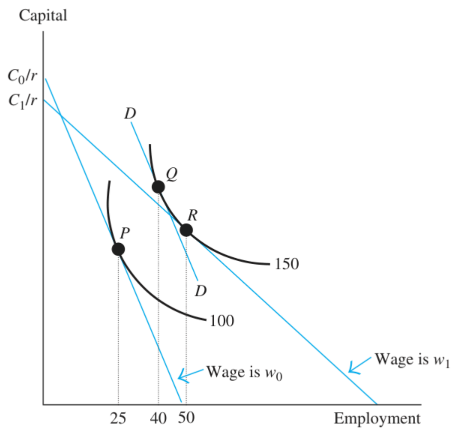
A. THE MECHANISM: WHAT HAPPENS WHEN w FALLS?

Suppose the wage drops from w_0 to w_1. The firm moves from initial equilibrium **P** to new equilibrium **R**. This movement is decomposed into two distinct effects.

1. The Scale Effect ($P \to Q$)

- **Logic:** A wage cut lowers the **Marginal Cost (MC)** of production.
- **Chain Reaction:** $w \downarrow \implies MC \downarrow \implies$ Firm expands output ($q \uparrow$).
- **Input Response:** To produce more output, the firm hires **more Labor** and uses **more Capital** (assuming both are "normal inputs").
- **Direction: Increases Employment.**

FIGURE 3-12 **Substitution and Scale Effects**
A wage cut generates substitution and scale effects. The scale effect (the move from point P to point Q) encourages the firm to expand, increasing the firm's employment. The substitution effect (from Q to R) encourages the firm to use a more labor-intensive method of production, further increasing employment.



2. The Substitution Effect ($Q \to R$)

- **Logic:** Labor is now relatively cheaper than Capital (w/r has decreased).
- **Chain Reaction:** The firm changes its production technique to become more **labor-intensive**, holding output constant.
- **Input Response:** The firm substitutes **Labor for Capital**.

- **Direction: Increases Employment** (but decreases Capital).

B. NET EFFECTS ON INPUTS

Input	Substitution Effect	Scale Effect	Net Total Effect
Labor (E)	Increases (↑)	Increases (↑)	**Unambiguously Increases**
Capital (K)	Decreases (↓)	Increases (↑)	**Ambiguous** (Depends on strength of effects)

- *Note on Capital:* If the Scale Effect > Substitution Effect, the firm hires and more capital (Complements). If Substitution > Scale, the firm hires but less capital (Substitutes).

B-2. EMPIRICAL ESTIMATES OF LABOR DEMAND ELASTICITY

Despite the theoretical clarity, estimating the actual magnitude of δ is empirically difficult (see Section 3-12 for methodology). The range of estimates from the literature is wide, but the main findings are:

- **Direction:** Virtually all studies confirm the demand curve is **downward sloping** ($\delta < 0$).
- **Short-Run Elasticity:** Closer to zero (capital is fixed, firm has limited flexibility).
- **Long-Run Elasticity:** Larger in magnitude (firm can adjust both inputs).
- **Typical Range:** Most estimates place the long-run own-wage elasticity of labor demand between -0.15 **and** -0.75, with a commonly cited benchmark around -0.3 **to** -0.4.

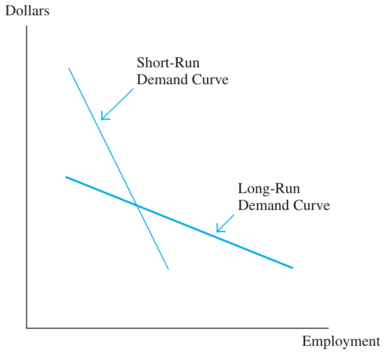
> **Exam Note**
>
> You do not need to memorise precise estimates. The key point is: (a) the curve is definitively downward sloping, and (b) long-run elasticity is **larger in magnitude** than short-run, consistent with the Le Chatelier Principle.

C. LONG-RUN VS. SHORT-RUN ELASTICITY

- **Short Run:** Capital is fixed. The firm can only respond via the Scale Effect (expanding output slightly) and diminishing marginal returns.
- **Long Run:** Capital is variable. The firm can respond via Scale **plus** Substitution (rearranging the factory floor to use more workers and fewer machines).
- **Le Chatelier Principle:** The firm has more flexibility in the long run. Therefore:

> **The Long-Run Labor Demand Curve is FLATTER (More Elastic) than the Short-Run Demand Curve.**

FIGURE 3-13 **The Short- and Long-Run Demand Curves for Labor**
In the long run, the firm can take full advantage of the economic opportunities introduced by a change in the wage. As a result, the long-run demand curve is more elastic than the short-run demand curve.



📝 EXAM PREPARATION NOTES

Likely 10-15 Mark Questions:

1. *Decompose the impact of a wage increase on the demand for labor into Substitution and Scale effects using Isoquant/Isocost analysis.* (This is the standard "Figure 3-12" question).
2. *Why is the long-run demand for labor more elastic than the short-run demand?*
3. *Explain why profit maximization implies cost minimization, but cost minimization does not necessarily imply profit*

maximization.

Diagram Checklist:

- **Figure 3-8 (Cost Minimization):** Tangency of Isoquant and Isocost.

Figure 3-12 (The Decomposition — CRITICAL):

Setup: Wage falls from w_0 to w_1. Firm moves from **P** (25 workers, 100 units output) to **R** (50 workers, 150 units output). The total movement $P \to R$ is decomposed using a **hypothetical isocost line DD**.

The DD Line (Key Construction Step):

- Draw an isocost line with the **new slope** ($-w_1/r$, i.e., flatter) but tangent to the **new higher isoquant** (150 units).
- The tangency point of DD with the new isoquant is **Point Q** (40 workers).
- DD is a *hypothetical* line — it is never actually observed. Its purpose is purely to isolate the two effects.
  Decomposition:

Move	Name	What is held constant	Direction of Employment
P $\to$ Q	Scale Effect	Input price ratio (slope = w_0/r)	Increases (25 $\to$ 40)
Q $\to$ R	Substitution Effect	Output level (stays on 150-unit isoquant)	Increases (40 $\to$ 50)

- **Scale Effect ($P \to Q$):** Move to a *higher* isoquant (firm expands output because MC fell), but measure this using the *old* price ratio (DD is parallel to the original isocost). This isolates the pure output-expansion component.
- **Substitution Effect ($Q \to R$):** Slide *along* the new isoquant from Q to R. Output is held constant but the firm adopts a more labour-intensive technique because labour is now cheaper relative to capital.

> **Capital Effect**
>
> As drawn in Figure 3-12, the firm uses **more capital** at R than at P — the Scale Effect (which increases capital demand) **outweighs** the Substitution Effect (which reduces capital demand). If Substitution > Scale, the firm would use *less* capital overall.

Drawing Sequence for Exam:
1. Draw original isoquant ($q = 100$), original isocost (slope $-w_0/r$), tangent at **P** (25 workers).
2. Draw higher isoquant ($q = 150$) above it.
3. Draw **DD**: flatter slope ($-w_1/r$), tangent to the *higher* isoquant at **Q** (40 workers).
4. Draw the *actual* new isocost (same slope as DD, but shifted to pass through the firm's new cost level), tangent to the higher isoquant at **R** (50 workers).
5. Label: $P \to Q$ = Scale Effect; $Q \to R$ = Substitution Effect; $P \to R$ = Total Effect.

SECTION 3-5: THE ELASTICITY OF SUBSTITUTION

📜 THESIS STATEMENT:

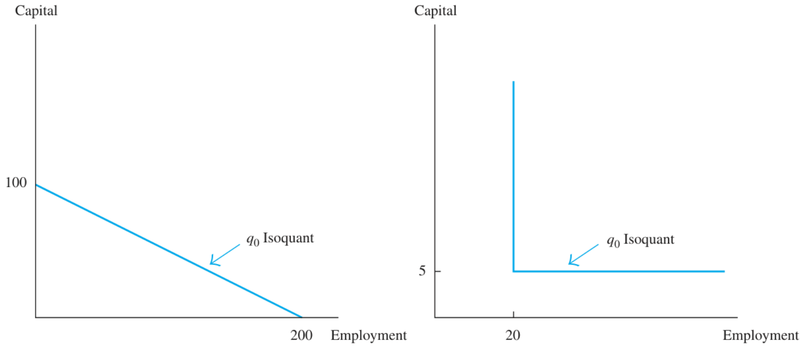
> **The magnitude of the substitution effect depends entirely on the curvature of the isoquant.** The Elasticity of Substitution measures the ease with which a firm can replace capital with labor (or vice versa) while keeping output constant.

A. THE CURVATURE OF ISOQUANTS

The shape of the isoquant reflects the technological possibilities of production.

1. **Perfect Substitutes (Linear Isoquants):**
   - **Shape:** Straight downward-sloping line.
   - **Logic:** Inputs are interchangeable at a constant rate (e.g., two red pencils replace one blue pencil).
   - **Implication:** A small change in input prices can cause a massive shift from all capital to all labor. The Substitution Effect is huge.

FIGURE 3-14 **Isoquants When Inputs Are Either Perfect Substitutes or Perfect Complements**
Capital and labor are perfect substitutes if the isoquant is linear (so that two workers can always be substituted for one machine). The two inputs are perfect complements if the isoquant is right-angled. The firm then gets the same output when it hires 5 machines and 20 workers as when it hires 5 machines and 25 workers.



(*a*) Perfect Substitutes (*b*) Perfect Complements

[7] Note that our definition of perfect substitution does *not* imply that the two inputs have to be exchanged on a one-to-one basis; that is, one machine hired for each worker laid off. Our definition implies only that the rate at which capital can be replaced for labor is constant.

2. **Perfect Complements (Right-Angled / Leontief Isoquants):**
   - **Shape:** L-shaped.
   - **Logic:** Inputs must be used in fixed proportions (e.g., one driver for one taxi).
   - **Implication:** No substitution is possible. Even if wages drop to zero, the firm cannot hire more labor without more capital. The Substitution Effect is zero.

B. THE ELASTICITY OF SUBSTITUTION (σ)

A numerical measure of the curvature.

$$\sigma = \frac{\%\Delta(K/E)}{\%\Delta(w/r)}$$

- **Interpretation:** The percentage change in the capital/labor ratio resulting from a 1% change in relative prices.
- **Values:**
  - $\sigma = \infty$: Perfect Substitutes (Flat isoquant).
  - $\sigma = 0$: Perfect Complements (Right-angled isoquant).
  - $\sigma > 0$: Standard convex isoquants (Cobb-Douglas).

Policy Implication:

If σ is high, minimum wage hikes (which increase w/r) will cause massive job losses because firms can easily automate (swap E for K). If σ is low, employment is less sensitive to wage hikes.

SECTION 3-6: POLICY APPLICATION — AFFIRMATIVE ACTION

📜 THESIS STATEMENT:

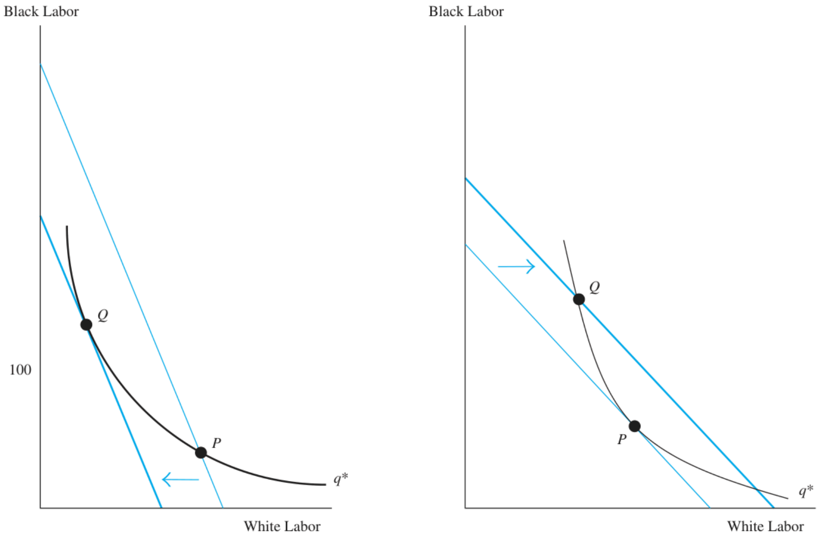
> **The economic efficiency of Affirmative Action depends critically on whether the market was initially discriminatory or color-blind.** If firms were discriminating (forgoing profits to avoid hiring black workers), Affirmative Action restores efficiency. If firms were color-blind (optimizing based on productivity), Affirmative Action introduces inefficiency.

A. THE MOD EL: TWO TYPES OF LABOR

Assume two inputs: Black Labor (E_B) and White Labor (E_W).

- **Isoquants:** Convex (Inputs are imperfect substitutes).
- **Isocosts:** Slope determines relative wages ($-w_W/w_B$).

FIGURE 3-15 Affirmative Action and the Costs of Production
(*a*) The discriminatory firm chooses the input mix at point *P*, ignoring the cost-minimizing rule that the isoquant be tangent to the isocost. An affirmative action program can force the firm to move to point *Q*, resulting in more efficient production and lower costs. (*b*) A color-blind firm is at point *P*, hiring relatively more whites because of the shape of the isoquants. An affirmative action program increases this firm's costs.



B. SCENARIO 1: THE DISCRIMINATORY FIRM

- **The Bias:** The firm dislikes hiring black workers. It ignores the market price ratio and acts as if black labor is more expensive than it actually is.
- **The Choice (Point P):** The firm operates **off** the tangency path, hiring too few black workers and too many white workers to minimize costs.
- **Affirmative Action Impact:**
  - The government mandates hiring more black workers.
  - This forces the firm *towards* the cost-minimizing tangency (Point Q).
  - **Result: Costs Fall.** The firm becomes *more* profitable because it stopped indulging its prejudice.

C. SCENARIO 2: THE COLOR-BLIND FIRM

- **The Reality:** The firm has no prejudice. It hires the optimal mix based purely on productivity and wages.
- **The Choice (Point P):** The firm is already at the cost-minimizing tangency.
- **Affirmative Action Impact:**
  - The government mandates hiring a different ratio (e.g., a quota).
  - This forces the firm *away* from the tangency (to Point Q).
  - **Result: Costs Rise.** The firm becomes *less* profitable because it is forced to use an inefficient input mix.

> 📋 Exam Takeaway
>
> - **Discriminatory Market:** AA → Efficiency Gain (Profits ↑).
> - **Competitive Market:** AA → Efficiency Loss (Profits ↓).

SECTION 3-7: MARSHALL'S RULES OF DERIVED DEMAND

📜 THESIS STATEMENT:

> **Alfred Marshall identified four laws that determine the Elasticity of Labor Demand.** Labor demand is more elastic (responsive to wage changes) when labor can be easily substituted, when the product market is competitive, when labor is a major cost, or when other inputs are elastic.

A. THE FOUR RULES

Labor demand is **more elastic** when:

1. **The Elasticity of Substitution (σ) is Greater:**

- *Mechanism:* If technology allows easy swapping of men for machines, a wage hike triggers a massive Substitution Effect.

2. **The Elasticity of Product Demand (η) is Greater:**
   - *Mechanism:* Wage $\uparrow \rightarrow$ Costs $\uparrow \rightarrow$ Prices $\uparrow$. If consumers are sensitive to price (High η), they stop buying. Output plummets $\rightarrow$ Employment plummets (Large Scale Effect).
3. **Labor's Share of Total Costs (S_L) is Greater:**
   - *Mechanism:* If labor is 80% of costs, a 10% wage hike raises total costs by 8% (huge). Prices must rise significantly, triggering a large Scale Effect.
   - *Contrast:* If labor is only 5% of costs, a wage hike barely affects final prices.
4. **The Supply Elasticity of Other Factors is Greater:**
   - *Mechanism:* If the firm wants to substitute capital for labor, it needs to buy capital. If capital prices skyrocket when demand rises (inelastic supply), the firm *cannot* afford to substitute. If capital supply is elastic, substitution is cheap and easy.

B. APPLICATION: UNIONS

Unions want **inelastic** labor demand so they can raise wages without causing unemployment.

- *Strategy:*
  - Limit substitutes (oppose automation/imports) $\rightarrow$ Lowers Rule 1 & 2.
  - Organize "craft" trades (electricians) where labor share is low $\rightarrow$ Lowers Rule 3.

SECTION 3-8: FACTOR DEMAND WITH MANY INPUTS

📜 THESIS STATEMENT:

> **In the real world, production uses multiple inputs (Skilled Labor, Unskilled Labor, Capital).** The relationship between these inputs is defined by the **Cross-Elasticity of Factor Demand**, which reveals whether inputs are **Substitutes** or **Complements**.

A. CROSS-ELASTICITY FORMULA

$$\delta_{ij} = \frac{\%\Delta x_i}{\%\Delta w_j}$$

- How does the demand for Input i change when the price of Input j changes?

B. SUBSTITUTES VS. COMPLEMENTS

1. **Substitutes ($\delta_{ij} > 0$):**
   - If price of Unskilled Labor (w_u) $\uparrow$, firm hires *more* Skilled Labor (E_s).
   - *Example:* Automated kiosks vs. Cashiers.
2. **Complements ($\delta_{ij} < 0$):**
   - If price of Capital (r) $\uparrow$, firm hires *less* Skilled Labor (E_s).
   - *Inputs "go together."*

C. THE CAPITAL-SKILL COMPLEMENTARITY HYPOTHESIS

- **Empirical Fact:** Data suggests that **Capital (K) and Skilled Labor (E_s) are Complements**, while **Capital (K) and Unskilled Labor (E_u) are Substitutes.**
- **Policy Consequence:** An investment tax credit (making K cheaper) will:
  - Increase demand for machines (K).
  - Increase demand for Skilled Labor (E_s).
  - Decrease demand for Unskilled Labor (E_u).
  - **Result:** Widening wage inequality.

Cross-Elasticity (3-8)

- δ > 0 = Substitutes, δ < 0 = Complements
- Capital + Skilled Labor = **Complements** | Capital + Unskilled Labor = **Substitutes**
- Cheaper capital → more machines + more skilled workers + fewer unskilled workers → **wage inequality widens**

Minimum Wage (3-10) — highest exam priority

- Standard model: binding minimum wage → employment ↓ (firms hire less) + labour supply ↑ (more workers enter) = **unemployment gap**
- Two-sector model: displaced workers flood uncovered sector → wages there **fall** (unless workers *queue* for covered jobs → wages in uncovered sector **rise** — Harris-Todaro logic)
- **Card & Krueger (1994):** NJ vs PA natural experiment (DiD) — employment didn't fall → hints at **monopsony**
- Non-compliance = firms just pay below minimum, get caught = only back-pay. Acts as a buffer → actual unemployment smaller than model predicts

---

Adjustment Costs (3-11)

- Variable costs → slow, smooth adjustment | Fixed costs → lumpy, sudden adjustment + **labour hoarding**
- Firing costs (EPL) → firms scared to hire in booms → dampens fluctuations but lowers average employment
- L = M × H. High fixed costs per worker → firm prefers **overtime**. High overtime premium → firm prefers **hiring new workers** (job sharing)

IV / Rosie the Riveter (3-12)

- Identification problem: can't tell if a data point shift is supply or demand moving
- Need an instrument that shifts **supply only**, not demand
- WWII draft = instrument. High draft state → more women entered workforce (supply shift) → traces out demand curve
- Result: female labour demand elasticity ≈ **-1.0** (unit elastic)

One table worth memorising:

Scenario	Employment	Wage
Covered sector	↓	↑
Uncovered sector (displaced workers)	↑	↓
Uncovered sector (queuing)	↓	↑
Monopsony	↑	↑

That's honestly everything you need. Card & Krueger + the two-sector model diagram are the most likely 10-markers.

SECTION 3-9: OVERVIEW OF LABOR MARKET EQUILIBRIUM

📜 THESIS STATEMENT:

> **The labor market clears where the Market Labor Supply curve intersects the Market Labor Demand curve.** This equilibrium balances the conflicting desires of workers (high wages) and firms (low wages).

A. THE EQUILIBRIUM MECHANISM

- **Equilibrium (E^*, w^*):** Quantity Supplied = Quantity Demanded.
- **Disequilibrium Adjustment:**
  - **If $w > w^*$:** Excess Supply (Unemployment). Workers compete for jobs, bidding wages **down**.
  - **If $w < w^*$:** Excess Demand (Labor Shortage). Firms compete for workers, bidding wages **up**.

B. EFFICIENCY

- In a perfectly competitive market without distortions (taxes, minimum wages), the equilibrium allocation maximizes the total gains from trade (Worker Surplus + Producer Surplus).

- Any deviation creates **Deadweight Loss**.

📝 EXAM PREPARATION NOTES

Possible 5-10 Mark Questions:

1. *State Marshall's Four Rules of Derived Demand. Explain why a high share of labor in total costs makes labor demand more elastic.*
2. *Using an Isoquant/Isocost diagram, show how Affirmative Action reduces costs for a discriminatory firm but increases costs for a color-blind firm.*
3. *Define the Capital-Skill Complementarity Hypothesis. What does it imply about the effect of cheaper technology on the wage gap between skilled and unskilled workers?*

Diagram Checklist:

- **Figure 3-14:** Draw Linear Isoquants (Substitutes) vs. L-Shaped Isoquants (Complements).
- **Figure 3-15 (Affirmative Action):**
  - Draw Isoquant and Isocost tangent at **Q** (Optimal/Color-blind).
  - Mark Point **P** off-tangency (Discriminatory).
  - Show that **Q** is on a lower isocost line than **P**.
- **Figure 3-16 (Shift vs Movement):**
  - Show how a change in the price of Capital shifts the demand curve for Labor (Up for substitutes, Down for complements).

SECTION 3-10: POLICY APPLICATION — THE EMPLOYMENT EFFECTS OF MINIMUM WAGES

📜 THESIS STATEMENT:

> **The imposition of a minimum wage acts as a price floor in the labor market.** In a perfectly competitive model, if the minimum wage is binding (set above equilibrium), it unambiguously creates unemployment by simultaneously reducing labor demand (firms hire fewer workers) and increasing labor supply (more workers enter the market). However, empirical evidence is mixed, leading to debates over the elasticity of labor demand.

A. THE STANDARD COMPETITIVE MODEL

Assume a single competitive labor market covered by the legislation.

1. **The Equilibrium:**
   - Initial Wage: w^*
   - Initial Employment: E^*
2. **The Policy Shock:**
   - Government sets Minimum Wage $\bar{w} > w^*$.
3. **The Demand Response (Displacement):**
   - Firms move **up** the labor demand curve.
   - Marginal Cost of labor > VMP_E.
   - Employment falls to $\bar{E}$.
   - **Displacement Effect:** $E^* - \bar{E}$ workers lose jobs.
4. **The Supply Response (Entry):**
   - Workers move **up** the labor supply curve.
   - Higher wage attracts new entrants (e.g., teenagers leaving school).
   - Labor Force grows to E_S.
5. **The Unemployment Gap:**

$$\text{Unemployment} = \text{Quantity Supplied}(E_S) - \text{Quantity Demanded}(\bar{E})$$

   - *Note:* Unemployment is caused *both* by firing existing workers and by new entrants failing to find jobs.

B. THE TWO-SECTOR MODEL (Covered vs. Uncovered)

In reality, not all jobs are covered by minimum wage laws (e.g., informal sector, agriculture, small firms).

1. **The Mechanism:**
    - **Sector 1 (Covered):** Wage rises to $\bar{w}$. Employment falls. Displaced workers lose jobs.
    - **Sector 2 (Uncovered):** Displaced workers migrate here to find work.
2. **The Supply Shift:**
    - The supply curve in the Uncovered Sector shifts **outward (Right)**.
    - **Result:** Wages in the Uncovered Sector **fall** below the original competitive level ($w_{uncovered} < w^*$).
3. **Alternative Scenario (Queuing):**
    - If workers prefer to "wait" for a high-paying job in the Covered Sector rather than work for peanuts in the Uncovered Sector, they may quit Uncovered jobs.
    - Supply in Uncovered shifts **Left**. Uncovered wages **rise**.
    - *Harris-Todaro Model Logic:* Migration stops when Expected Wage (Covered) = Actual Wage (Uncovered).

C. EMPIRICAL EVIDENCE & CONTROVERSY

1. The "Old" Consensus

- Time-series studies focused on teenagers.
- **Elasticity Estimate:** Between **-0.1 and -0.3**.
- *Meaning:* A 10% hike in minimum wage reduces teen employment by 1-3%.

2. The New Wave: Card & Krueger (1994)

- **The Natural Experiment:** New Jersey raised minimum wage ($4.25 $\rightarrow$ $5.05). Pennsylvania did not.
- **Method: Difference-in-Differences (DiD)**.
    - Measured employment in fast-food restaurants in NJ (Treatment) vs. PA (Control).
- **The Finding:** Employment in NJ actually **increased** (or stayed constant) relative to PA.
- **Implication:** Labor demand might not be perfectly competitive (possible **Monopsony** power, where firms have some control over wages).

> **Exam Note: The "Revisionist" View**
>
> If specific case studies show zero or positive employment effects, it challenges the standard competitive model. It suggests firms might have **Monopsony Power** (discussed in Ch 4) or that efficiency wage gains offset the cost.

D. COMPLIANCE WITH THE MINIMUM WAGE

- **The Reality:** Not all firms comply with minimum wage legislation.
- **Scale of Non-Compliance:** In 2010 (federal minimum = $7.25/hr), approximately **2.5 million workers (3.5% of all hourly workers)** were paid below the statutory minimum.
- **Why Non-Compliance Is Rational for Firms:**
    - When a violation is *detected*, the penalty is merely a **back-pay settlement** — the firm pays workers the shortfall for the last two years of work.
    - Punitive damages are rare.
    - **Effect:** A firm breaking the law and getting caught receives an effective **interest-free loan** on the unpaid portion of its wage bill for up to two years.
    - A firm breaking the law and *not* getting caught (the majority of cases) simply continues paying the lower competitive wage indefinitely.
- **Macroeconomic Implication:** The greater the degree of non-compliance, the **smaller** the employment reduction caused by the minimum wage and the **lower** the resulting unemployment rate.

> **Key Insight**
>
> Non-compliance acts as a partial circuit-breaker on the standard model's unemployment prediction. It is one reason why observed employment effects of minimum wage hikes are smaller than the simple competitive model predicts.

E. THE LIVING WAGE

- **Definition:** "Living wage" ordinances are local laws (enacted by cities/municipalities) setting minimum wages **far above** the federal minimum. They typically cover **municipal employees** or workers at firms with government contracts.
- **Scale:** By the early 2000s, approximately **100 US cities** had enacted such ordinances.
- **Why Employment Effects Are Hard to Detect:**
  1. **Narrow Coverage:** Very few workers fall under living wage laws, making it statistically difficult to identify an impact.
  2. **Selection Problem:** Cities that choose to enact living wage laws may have systematically different labour market conditions than cities that do not — making the "control group" unclear.
- **Empirical Status:** Given these identification challenges, the evidence on living wage employment effects remains limited and inconclusive relative to the broader minimum wage literature.

SECTION 3-11: ADJUSTMENT COSTS AND LABOR DEMAND

📜 THESIS STATEMENT:

> **Labor is not a perfectly variable factor; changing the level of employment incurs Adjustment Costs.** These costs create "friction," causing firms to adjust employment slowly (lagging behind shocks) and creating a distinction between the number of workers (bodies) and the number of hours worked.

A. TYPES OF ADJUSTMENT COSTS

1. Variable Adjustment Costs

- **Definition:** Costs that change with the *rate* of adjustment (e.g., training costs are higher if you hire 100 people in one day vs. 100 people over a year).
- **Implication:** Firms adjust employment **slowly** and smoothly to smooth out these costs.
- **The Path:** If demand jumps, employment moves asymptotically toward the new equilibrium (Path A $\rightarrow$ B in Figure 3-22).

2. Fixed Adjustment Costs

- **Definition:** A lump-sum cost incurred regardless of how many workers are hired/fired (e.g., administrative setup, legal fees for firing).
- **Implication:** Employment changes are **lumpy** and **sudden**. Firms do not change employment for small shocks; they wait until the shock is large enough to justify the fixed cost (Labor Hoarding).

B. ASYMMETRIC COSTS (Firing Costs)

- **Context:** Employment Protection Legislation (EPL) in Europe mandates high severance pay.
- **Effect:** Firing costs > Hiring costs.
- **Result:** Firms are terrified to hire during booms because they know they cannot fire during busts.
- **Outcome:** Dampens employment fluctuations but lowers overall average employment.

C. THE TRADE-OFF: WORKERS VS. HOURS

Firms can increase labor input (L) by hiring more **Workers (M)** or increasing **Hours (H)**.

$$L = M \times H$$

- **Fixed Costs of Employment (F):** Health insurance, training, recruitment. These are paid *per body*.
- **Variable Costs (w):** Overtime pay. This is paid *per hour*.
- **The Optimization:**
  - **High Fixed Costs ($F \uparrow$):** It is expensive to hire new bodies. Firm prefers to pay **Overtime** to existing staff.
  - **High Overtime Premium ($w_{OT} \uparrow$):** It is expensive to work long hours. Firm prefers to hire **New Workers** (Job Sharing).

> 📋 **Policy Implication: Work Sharing**
>
> Policies that mandate shorter workweeks (e.g., France's 35-hour week) aim to force firms to substitute **Workers for Hours**.
>
> - *Success Condition:* Depends on whether the fixed cost of hiring (F) is low enough. If F is high, firms may just shrink output rather than hire new workers.

SECTION 3-12: ROSIE THE RIVETER AS AN INSTRUMENTAL VARIABLE

📜 THESIS STATEMENT:

> **Estimating the elasticity of labor demand is difficult due to the "Identification Problem": supply and demand usually shift simultaneously.** To isolate the slope of the demand curve, economists use **Instrumental Variables (IV)**—shocks that shift the Labor Supply curve without affecting Labor Demand.

A. THE IDENTIFICATION PROBLEM

- **The Trap:** We observe real-world data points (w_1, E_1) and (w_2, E_2).
- **The Confusion:** Did we move from 1 to 2 because Demand shifted? Or Supply shifted? Or both?
  - If *both* shifted, connecting the dots creates a meaningless line ("Curve ZZ"), not a demand curve.
- **The Goal:** We need to trace out the **Demand Curve**. To do this, we need the Demand curve to stay still while the **Supply curve moves**.

B. THE INSTRUMENTAL VARIABLE (IV)

An **Instrument** is a variable that:

1. **Relevance:** Causes a significant shift in Labor Supply.
2. **Exogeneity:** Is uncorrelated with shifts in Labor Demand (does not directly affect firm productivity).

C. CASE STUDY: WWII MOBILIZATION (Acemoglu, Autor, Lyle)

- **The Context:** World War II drafted millions of men. Women ("Rosie the Riveter") entered the workforce to replace them.
- **The Instrument: State Mobilization Rate.**
  - Different states had different percentages of men drafted (e.g., farmers were exempted, so agricultural states had low drafts; industrial states had high drafts).
  - High Draft $\rightarrow$ Low Male Supply $\rightarrow$ High Female Entry (Supply Shift).
- **The Logic:**
  - The Draft shifted Female Labor Supply (Right).
  - The Draft *did not* change female productivity (Labor Demand).
  - Therefore, comparing wage/employment changes across states traces out the **Labor Demand Curve**.
- **The Result:**
  - States with high mobilization saw huge increases in female employment and decreases in female wages (relative to trend).
  - **Estimated Elasticity:** $\delta \approx -1.0$ (The long-run demand for female labor was unit elastic).

> 🔥 **Exam Note: Why this matters**
>
> This section teaches **methodology**. In an exam, if asked "How do economists estimate elasticity?", do not just give the formula. Explain that they need an **exogenous supply shock** (like the Draft) to distinguish the demand curve from the supply curve.

📝 COMPARATIVE TABLE: MINIMUM WAGE IMPACTS

Scenario	Employment Effect	Wage Effect
Competitive Model (Covered)	**Decreases** $(E \downarrow)$	**Increases** $(\bar{w})$
Competitive Model (Uncovered)	**Increases** (Supply shift right)	**Decreases** (Supply shift right)
Monopsony Model (New Wave)	**Increases** (potentially)	**Increases**
Total Labor Market	**Unemployment** $(E_S - E_D)$	Average wage depends on coverage

---

📝 EXAM PREPARATION NOTES

Likely 10-15 Mark Questions:

1. *Using a diagram, analyze the effect of a minimum wage on the covered and uncovered sectors of a dual labor market. Under what conditions might wages in the uncovered sector rise?* (Answer: Queueing/Harris-Todaro).
2. *Explain the "Identification Problem" in estimating labor demand curves. How did Acemoglu et al. use WWII mobilization rates to solve this?*
3. *Distinguish between fixed and variable adjustment costs. How do fixed costs explain "Labor Hoarding" during recessions?*

Diagram Checklist:

- **Figure 3-19:** Standard Minimum Wage (Triangle of unemployment).
- **Figure 3-20 (Two-Sector Model):**
  - Panel A (Covered): Wage floor, Employment falls.
  - Panel B (Uncovered): Supply shifts Right (Displaced workers arrive) $\rightarrow$ Wage falls.
- **Figure 3-22 (Adjustment Costs):** Show the smooth/asymptotic path of employment adjustment towards equilibrium.
- **Figure 3-23 (Identification):** Show a mess of points (Z-Z) vs. a clean Demand curve traced by shifting Supply curves.

📋 EXECUTIVE SUMMARY (Key Concepts from Text)

- **Short-Run Rule:** Firms hire until Wage (w) = Value of Marginal Product (VMP_E).
- **Long-Run Rule:** Firms choose input mix where the ratio of marginal products equals the ratio of input prices ($MP_E/MP_K = w/r$).
- **Wage Decline:** Triggers **Substitution Effect** (swap capital for labor) and **Scale Effect** (expand output). Both increase employment.
- **Elasticity:** Long-run demand is **more elastic** than short-run demand (Le Chatelier Principle).
- **Capital-Skill Complementarity:** Capital substitutes for unskilled labor but complements skilled labor.
- **Minimum Wage:** Creates unemployment in competitive markets by inducing displacement (firing) and entry (supply increase).
- **Adjustment Costs:** Variable costs cause slow/smooth employment adjustment; Fixed costs cause lumpy/sudden adjustment.

❓ QUESTION 1: OPTIMIZATION LOGIC

Text Source: Review Q1, Q4; Problem 3-7, 3-10.

📜 THESIS:

> **Profit maximization requires alignment between the cost of an input and the revenue it generates.** In the short run, this is the $w = VMP_E$ condition. In the long run, cost minimization ($MP_E/w = MP_K/r$) is a necessary *but insufficient* condition for profit maximization.

A. THE SHORT-RUN CONDITION ($w = VMP_E$)

- **The Logic:**
  - If $VMP_E > w$: The last worker brings in more revenue than they cost. **Hire more.**
  - If $VMP_E < w$: The last worker costs more than they generate. **Fire them.**
  - **Equilibrium:** Hiring stops exactly when the cost equals the return.
- **Equivalence to $P = MC$:**
  - Producing one extra unit requires $1/MP_E$ hours of labor.
  - Marginal Cost (MC) = $w \times (1/MP_E)$.
  - If $P = MC$, then $P = w/MP_E$, which rearranges to $P \times MP_E = w$.

B. COST MINIMIZATION VS. PROFIT MAXIMIZATION

Scenario: A firm equates the ratio of marginal products to the ratio of prices:

$$\frac{MP_E}{MP_K} = \frac{w}{r}$$

Does this imply the firm is maximizing profits?

- **Answer: NO.**

- **Reasoning:** This condition only implies **Cost Minimization**. It means the firm is producing *some* level of output as cheaply as possible (being on the expansion path).
- **The Missing Link:** It does not tell the firm *how much* output to produce. To maximize profit, the firm must satisfy the condition AND ensure that $MC = MarginalRevenue$.

C. NUMERICAL APPLICATION (Problem 3-10)

Given: $w = 10$, $r = 25$, $P = 50$. Production: $q = E^{0.5}K^{0.5}$. Capital fixed at $K = 1600$.

1. **Calculate** MP_E: Derivative w.r.t E is $0.5(K/E)^{0.5}$.
   - With $K = 1600$: $MP_E = 0.5(1600/E)^{0.5} = 20E^{-0.5}$.
2. **Set Condition:** $w = P \times MP_E$.
   - $10 = 50 \times (20E^{-0.5})$.
   - $10 = 1000/\sqrt{E}$.
   - $\sqrt{E} = 100$.
   - $E = 10,000$. The firm hires 10,000 hours.

❓ QUESTION 2: WAGE CHANGES & ELASTICITY

Text Source: Review Q5, Q6; Problem 3-4, 3-13.

📜 THESIS:

> **A wage change triggers a dual response in the long run: shifting the production technique (Substitution) and changing the production volume (Scale).** The magnitude of the employment change depends on Marshall's Rules.

A. DECOMPOSING A WAGE INCREASE ($w \uparrow$)

1. **Substitution Effect:**
   - Labor is relatively more expensive (w/r rises).
   - Firm slides along the Isoquant.
   - **Result:** Labor ↓, Capital ↑.
2. **Scale Effect:**
   - Marginal Cost of production rises ($MC \uparrow$).
   - Firm produces less output ($q \downarrow$).
   - **Result:** Labor ↓, Capital ↓ (assuming normal inputs).
3. **Net Result:**
   - **Labor:** Unambiguously decreases (Both effects are negative).
   - **Capital:** Ambiguous (Substitution says buy more machines; Scale says shrink factory).

B. MARSHALL'S RULES (Why is demand elastic?)

Labor demand is **more elastic** if:

1. **Elasticity of Substitution is high:** Easy to replace men with machines.
2. **Product Demand is elastic:** Consumers quit buying if price rises.
3. **Labor Share is high:** Wages are a huge chunk of costs.
4. **Supply of other factors is elastic:** Easy to buy more machines.

- **Application (Problem 3-13):** Why is demand for **Nurses/Teachers** inelastic?
  - Mainly because **Product Demand is Inelastic**. People (or governments) do not stop consuming healthcare/education significantly when costs rise. Also, substitution possibilities are limited (low σ).

❓ QUESTION 3: MINIMUM WAGE IMPACTS

Text Source: Review Q8, Q9; Problem 3-9, 3-12.

📜 THESIS:

> **Minimum wages create a wedge between supply and demand.** In a covered sector, this generates unemployment. In a two-sector economy, the effects ripple into the uncovered sector, depressing wages there via migration.

A. CALCULATING UNEMPLOYMENT (Problem 3-9)

Given: Supply $E_S = 10 + w$. Demand $E_D = 40 - 4w$.

1. **Equilibrium (w^*):** Set $S = D$.
   - $10 + w = 40 - 4w \Rightarrow 5w = 30 \Rightarrow w^* = 6$.
   - Employment $E^* = 16$.
2. **Minimum Wage Shock ($\bar{w} = 8$):**
   - **Demand:** $40 - 4(8) = 8$ workers hired.
   - **Supply:** $10 + 8 = 18$ workers looking.
   - **Unemployment:** $18 - 8 = 10$ workers.
   - **Displacement:** $16 - 8 = 8$ lost jobs.
   - **New Entrants:** $18 - 16 = 2$ new job seekers.

B. COVERED VS. UNCOVERED SECTORS

If the minimum wage only applies to Sector A (Covered):

1. Displaced workers from Sector A lose jobs.
2. They migrate to Sector B (Uncovered).
3. Supply in Sector B shifts **Right**.
4. **Result:** Wages in the Uncovered sector **fall**.

❓ QUESTION 4: ADJUSTMENT COSTS

Text Source: Review Q11; Problem 3-14.

📜 THESIS:

> **Firms do not adjust labor instantly because hiring and firing is costly.** The *type* of cost determines the *speed* of adjustment.

A. THE MECHANISM

1. **Variable Costs:** Costs rise with the *number* of hires (e.g., training).
   - **Strategy:** Adjust slowly. Smooth the hiring over time.
2. **Fixed Costs:** Lump-sum cost regardless of number (e.g., legal fees, setup).
   - **Strategy:** "Lumpy" adjustment. Wait until the shock is big enough, then adjust all at once.

B. CASE STUDY: TRUCKERS VS. PROFESSORS (Problem 3-14)

- **Trucking Firm (Low Regulation):**
  - Likely **Variable Costs.**
  - **Adjustment:** Relatively fast and smooth transition to new employment levels.
- **Liberal Arts College (Tenure System):**
  - **High Fixed/Firing Costs:** Tenure makes firing nearly impossible or extremely expensive.
  - **Adjustment:**
    - *Expansion:* Very cautious hiring (fear of being stuck with tenured staff).
    - *Contraction:* Extremely slow (attrition only).
  - **Result:** Employment levels at the college will be "sticky" and unresponsive to short-term economic changes compared to the trucking firm.

❓ QUESTION 5: CAPITAL-SKILL COMPLEMENTARITY

Text Source: Review Q7; Problem 3-15.

📜 THESIS:

> **Technology is not neutral.** It interacts differently with different types of labor.

A. THE HYPOTHESIS

- **Unskilled Labor & Capital: Substitutes.** Machines replace ditch diggers and assembly line workers. ($P_K \downarrow \implies E_{unskilled} \downarrow$).
- **Skilled Labor & Capital: Complements.** Machines need engineers and programmers to run them. ($P_K \downarrow \implies E_{skilled} \uparrow$).

B. IMPLICATION FOR INEQUALITY

- If the price of capital falls (due to computers/tech progress):
  - Demand for skilled workers rises (Wages ↑).
  - Demand for unskilled workers falls (Wages ↓).
  - **Result:** The wage gap widens.

❓ QUESTION 6: INSTRUMENTAL VARIABLES (IV)

Text Source: Review Q12; Problem 3-6.

📜 THESIS:

> **To estimate the elasticity of labor demand, economists must isolate shifts in the Supply Curve.** Observing equilibrium points moving is insufficient because both curves usually shift.

A. THE IDENTIFICATION PROBLEM

- If wages rise and employment rises, is demand sloping up? No. It likely means Demand shifted right. We cannot see the *slope* of the demand curve if the demand curve itself is moving.

B. THE SOLUTION

- Find an **Instrument**: A variable that shifts Labor Supply but **does not** shift Labor Demand.
- **Example (WWII Mobilization):**
  - The draft removed men (Supply Shift Left).
  - It did not change the technology of factories (Demand constant).
  - **Result:** We trace out the Demand Curve by observing how wages and employment reacted to the supply shock.
- **Requirement:** The instrument must be correlated with supply but uncorrelated with the error term in the demand equation.